Group Work (Contact weekend) Assignment

Agenda

Friday:

15.00 Introduction of Group assignment and Creation of groups (4-5 students in a group)

16.00-20.00 **Group Work**

- Roles of group
- Do a schedule of project work
- Schedule a meetings (at least one per week)
- Do an EDA (you can use ChatGPT/Claude/CoPilot to do this)
- Preparation of dataset
    - Do you need the enrich data?
    - Do you need the information about the day of the week?
    - Do you need the feature engineering?
    - What weather information you need?
    - Think, what are the most relevant features in the dataset?
- Pre-study of Time serial neural network (TSNN)
    - Each student selects one TSNN algorithm to study and explore in more detail.
    - Do a study of TSNN
        - RNN, LSTM, GRUs, TCNs, Transformers, WaveNet, etc
    - Select the neural network
        - Do a documentation of your chosen type of neural network
        - Do an implemented
    - **Each student does an own neural network documentation and implementation**
    - **Each student write down how to use AI during the process**

Saturday:

9.00 – 12.00 Group work

- Group work continues…

12.30- Checking the state of assignment

- If you have completed everything, you are free to leave and go for a late lunch.

Assignment

The purpose of this exercise is to develop as accurate as possible Neural network based model that predicts electricity consumption in a property (the real data from the real house)

Goal(s):

- Predict the next hour electricity consumption

- Extra: Predict the next day (24h) electricity consumption

Dataset

The dataset description

The hourly consumption of the property for the period from 31.12.2015 22.00 pm. to 31.5.2022 20.00 pm.

- Measurement 31.12.2015 22.00 means the consumption between 21.00-22.00
- The hourly weather information for the period from 1.1.2017 to 31.5.2022

Zipped dataset file can be found from this zip file (Electricity_consumption.zip) in Moodle Datasets Tab:

- DATA: Dataset_mansion.csv
- Explanation of columns: Dataset_mansion.txt

Dataset splitting:

- Training dataset: 1.1.2016-30.9.2020
- Test dataset: 1.10.2020-30.9.2021

Evaluation datasets

Evaluate your implementation using appropriate loss function as follows

- Evaluation 1.10.2021-31.12.2021
- Evaluation 1.1.2022-28.2.2022
- Evaluation 1.3.2022-31.5.2022

Information about the property

Type: House

Type of heating: electric boiler, water pipes inside the floor.

Area: Apartment 138m2 + storage 10m2 + garage 30m2 (semi-heated)

Family: 2 adults, 3 children

Deadlines

Documentation should be returned to the Moodle return box Sunday 8.12.2026 at 23.59.

Documentation

The group will produce only one document, which will be done to the Jamk thesis template and project reporting instructions: https://help.jamk.fi/raportointiohje/en/

NOTE: All references must be done using Jamk APA7 😊

- Prestudy of Time serial neural network
- Short summary about the EDA
- Why selected implemented neural networks?
- Each student needs to do own implementation (code neural network)

- Do many different solutions/implementations
- Attach your neural network structure
- You can add code snipping to the document
- Share your implementation with your groupmate
- Results:
    - Collect results in one table
    - Explain the results
- Conclusion:
    - Is there some specific days in which the error is big?
        - If is, do you find the explanation
    - Are the accuracy same using Test, Eval1, Eval2 and Eval3 Datasets?
- Add jupyter notebook html files as attachment

Grading

Group work is evaluated 0-20 points based on:

- Group work evaluation (gwe) by Mika: 0-20
- Peer evaluation (*pe*) for each student and self evaluation (se): 0-5
    - Mean of *pe*
    - *Se*

How to calculate grade:

$$Points = \frac{Mean\ of\ pe}{5} * \frac{Se}{5} * gwe$$

Some example of evaluation

		Example 1		
Group Work				
	GW evaluation	Mean of peer eval	Self evaluation	Points
Student1	14	5	5	**14**
Student2	14	3	5	**8,4**
Student3	14	0	5	**0**

		Example 2		
Group Work				
	GW evaluation	Mean of peer eval	Self evaluation	Points
Student1	10	5	5	**10**
Student2	10	3	4	**4,8**
Student3	10	0	2	**0**

Notes:

- If you don't do anything *i.e.* mean of peer eval = 0 -> 0 points
- Group Work documentation (GW evaluation) is evaluated using Thesis assessment criteria